

3D Human Reference Body: Multiscale Exploration and Visualization of Biomolecular Data in Virtual Reality

Andreas Bueckle¹

¹ Department of Intelligent Systems Engineering, Luddy School of Informatics, Computing, and Engineering, Indiana University, Bloomington, IN, USA

Introduction: We would like to present an update on our ongoing Common Fund Data Ecosystem (CFDE) Pilot Project (R03) on the research and development of a novel, 3D, immersive, virtual reality (VR) exploration tool using a federated collection of datasets from three CFDE efforts: Human BioMolecular Atlas Program (HuBMAP, 469 datasets), Cellular Senescence Network (SenNet, 54 datasets), and Genotype-Tissue Expression (GTEx, 15 datasets), plus 124 datasets from other sources. The Aim of our work is to enable FAIR access to datasets from these three Common Fund programs via an immersive VR interface with embedded 2D/3D visualizations.

Method: Using existing, Linked Open Data (LOD) infrastructure built for the Human Reference Atlas (HRA), we federate cell type populations from datasets, 3D extraction sites, and anatomical structures across HuBMAP, SenNet, and GTEx in the HRA (Sub-Aim 1). Then, we retrieve existing cell type populations for these extraction sites and registered datasets given existing cell-by-gene matrices or proteomics markers (662 datasets in total as of February 2026). For the related Sub-Aim 2, we are deploying CFDE datasets in the HRA Organ Gallery in VR, an existing research prototype that allows the immersive exploration of this multiscale, multimodal, harmonized data. Building on our prior research on data visualization literacy, VR, and immersive analytics, we add 2D and 3D visualization capabilities to the HRA Organ Gallery to visualize cell type populations at the dataset, 3D extraction site, and anatomical structure (atlas) level. All cell type populations are connected to 3D registration sites inside one of 77 existing 3D reference organs (spatial), annotated with anatomical structure(s) that are aligned to the Uberon ontology (biological structure), and associated with clinical (donor) metadata, such as age, sex, race, etc.

Summary: We would like to demo a preview of these new features, both in a dedicated session but also informally throughout the CFDE Spring Meeting 2026 to gather feedback from stakeholders, incl. computational biologists, biomedical data visualization experts, and physician-scientists across different CFDE teams.

Conclusion: The app is available via the Meta Store

(<https://www.meta.com/experiences/5696814507101529>). A website on the HRA Organ Gallery is available at <https://humanatlas.io/hra-organ-gallery>. We will have a VR headset on hand for in-person demos.